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Math 400

Section 2.3 Calculating Limits Using the Limit Laws

Objective 1: Limits of Polynomial and Rational Functions.

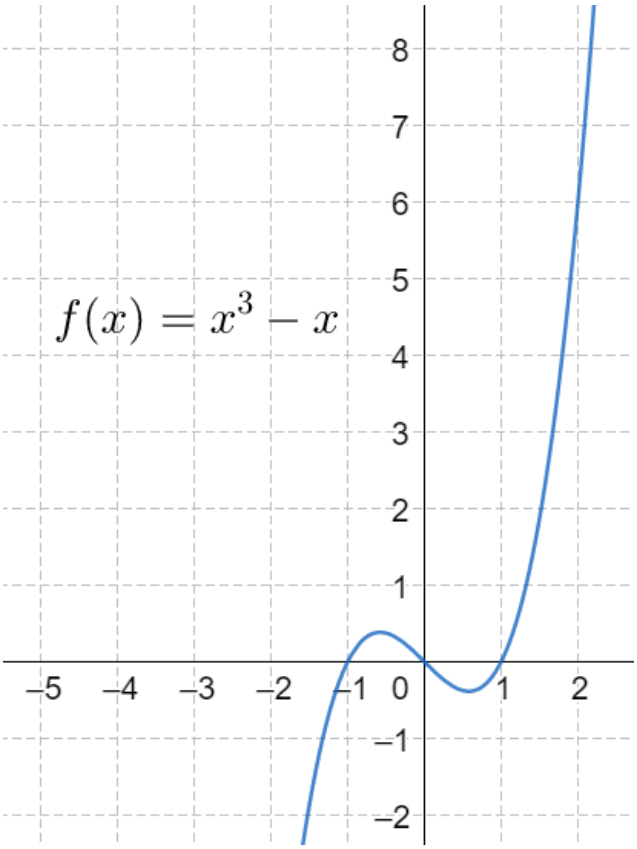
Direct Substitution Property If f is a polynomial or a rational function and a is in the domain of f, then

$$\lim_{x \to a} f(x) = f(a)$$

Graphs of polynomial functions and rational functions are continuous on their domains. You can find a limit by just evaluating the function at $x = a$. This is assuming that a is in the domain of the function.

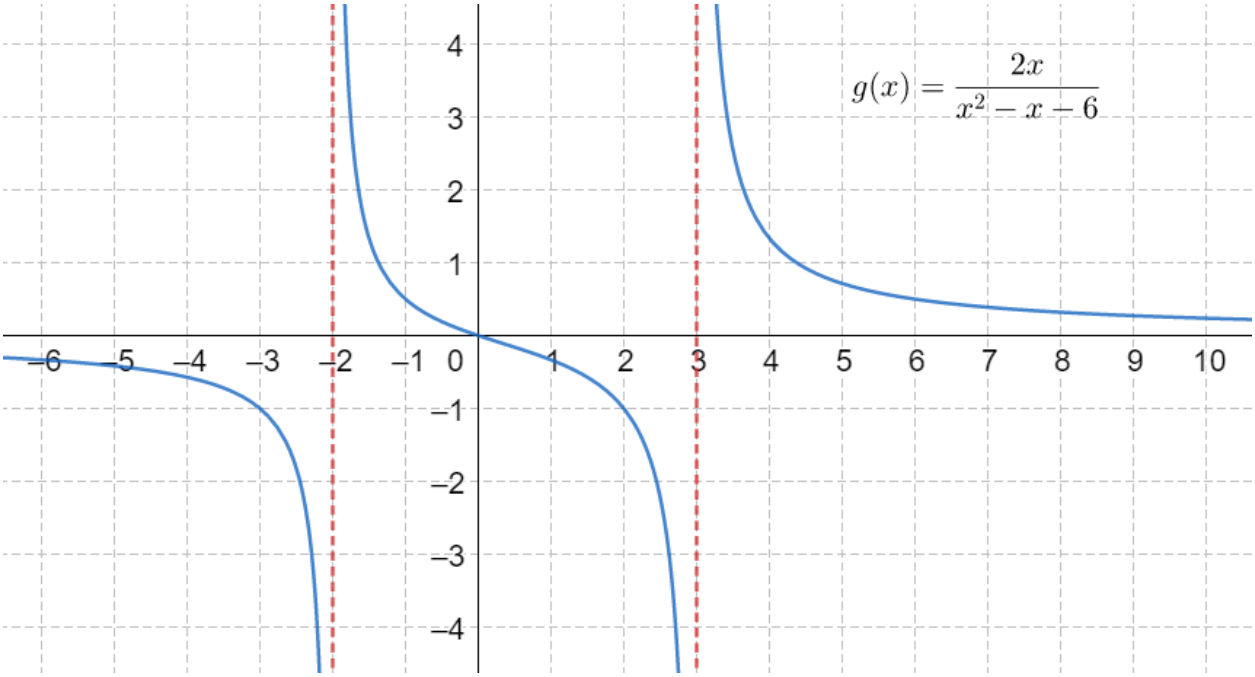
Example 1: Consider the polynomial function $f(x) = x^3 - x = x(x-1)(x+1)$.

Find $\lim_{x \to 2}(x^3 - x)$.



Example 2: Consider the rational function $g(x) = \frac{2x}{x^2-x-6} = \frac{2x}{(x-3)(x+2)}$.

Find $\lim_{x\to 0} \frac{2x}{x^2-x-6}$.



Example 3: Find the following limits.

$(a.)\ \lim_{x\to -1}(7x^4 - 9x^2 + 10)$

$(b.)\ \lim_{x\to 4}\dfrac{x^2 + 2x - 9}{x^3 + 1}$

Objective 2: Limits Laws.

Limit Laws

Suppose c is a constant and the limits $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist. Then

1.) $\lim_{x\to a}[f(x)+g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$

2.) $\lim_{x\to a}[f(x)-g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$

3.) $\lim_{x\to a}[cf(x)] = c\lim_{x\to a} f(x)$

4.) $\lim_{x\to a}[f(x)\cdot g(x)] = \lim_{x\to a} f(x)\cdot \lim_{x\to a} g(x)$

5.) $\lim_{x\to a}\dfrac{f(x)}{g(x)} = \dfrac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)} \qquad if \quad \lim_{x\to a} g(x) \neq 0$

6.) $\lim_{x\to a}[f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n \quad where\ n\ is\ a\ positive\ integer$

7.) $\lim_{x\to a} c = c$

8.) $\lim_{x\to a} x = a$

9.) $\lim_{x\to a} x^n = a^n \quad where\ n\ is\ a\ positive\ integer$

10.) $\lim_{x\to a} \sqrt[n]{x} = \sqrt[n]{a} \quad where\ n\ is\ a\ positive\ integer$

If n is even, we assume that $a > 0$.

11.) $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)} \quad where\ n\ is\ a\ positive\ integer$

If n is even, we assume that $\lim_{x\to a} f(x) > 0$.

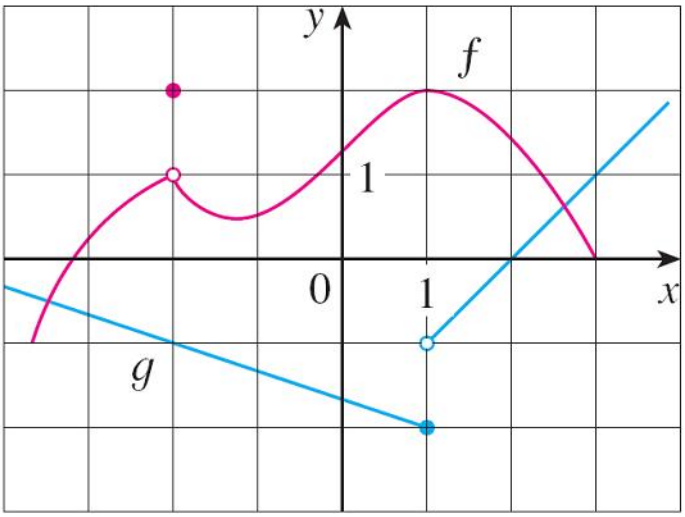
Example 4: Find the following limits.

$(a.)\ \lim_{x\to 3}\left(7x+1-\frac{x}{4x+2}\right)$

$(b.)\ \lim_{x\to 4}\sqrt{\frac{x^2+2x-9}{x^3+1}}$

$(c.)\ \lim_{x\to -1}(7x^4-9x^2+10)^6$

Example 5: Use the Limit Laws and the graphs of f and g to evaluate the following limits, if they exist.



$(a.)$ $\lim_{x\to 2}[f(x)+3g(x)]$

$(b.)$ $\lim_{x\to -2}\dfrac{f(x)}{[g(x)]^2}$

Objective 3: Indeterminate Form: $\frac{0}{0}$.

If $f(x) = g(x)$ when $x \neq a$, then $\lim_{x \to a} f(x) = \lim_{x \to a} g(x)$, provided the limits exist.

Example 6: Find the following limit, if it exists. (We will do this algebraically.)

$$\lim_{x \to 8} \frac{x^2 - 9x + 8}{x - 8}$$

Example 7: Find the following limit, if it exists. (We will do this algebraically.)

$$\lim_{x \to 4} \frac{x^2 - 16}{x^2 - 6x + 8}$$

Example 8: Find the following limit, if it exists. (We will do this algebraically.)

$$\lim_{x \to \pi} \frac{\cos^2 x + 3\cos x + 2}{\cos^2 x - 1}$$

Example 9: Find the following limit, if it exists. (We will do this algebraically.)

$$\lim_{x \to -1} \frac{\sqrt{2 - 7x} - 3}{x + 1}$$

Limits with Absolute Values

$f(x) = |x|$

Example 10: Find the following limit, if it exists. (We will do this algebraically.)

$$\lim_{x \to 0} \frac{|x|}{x}$$

Example 11: Find the following limit, if it exists. (We will do this algebraically.)

$$\lim_{x \to 4} \frac{|x-4|}{x-4}$$

Example 12: Find the following limit, if it exists. (We will do this algebraically.)

$$\lim_{x \to -6} \frac{x^2 + 4x - 12}{|x + 6|}$$

Objective 4: The Squeeze Theorem.

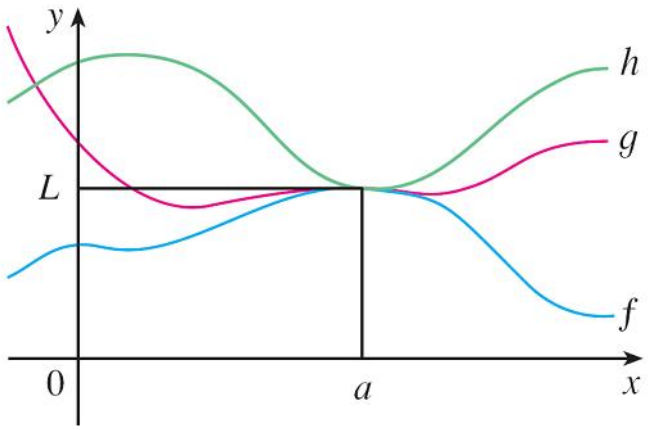
3 The Squeeze Theorem If $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$$

then

$$\lim_{x \to a} g(x) = L$$

It says that if g (x) is squeezed between f (x) and h (x) near a, and if f and h have the same limit L at a, then g is forced to have the same limit L at a.



Example 13: Find the following limit.

$$\lim_{x \to 0} x^2 \cos \frac{1}{x}$$

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Example 13: (continued)

Graph of $y = x^2 \cos\frac{1}{x}$.

